

Pytorch and Scikit-learn

Network Intrusion Detection Lab

Machine Learning and Security

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Scikit-learn and PyTorch Lab

Network Intrusion Detection

NSL-KDD Dataset

Dataset: Use the NSL-KDD dataset, for network intrusion detection.

- features: 41 attributes, including connection duration, protocol type, data bytes, and login attempts
- labels: Classify each record as "normal traffic" or "attack.
- includes multiclass labels specifying the attack

Data split:

- training set: 125,973 records
- test set: 22,544 records

NSL-KDD Dataset

Dataset: Use the NSL-KDD dataset, for network intrusion detection.

- categorical features: ['protocol_type', 'service', 'flag']
- categorical labels: ['attack']
- columns to drop: 'attack score'
  - indicates difficulty of the attack. We don’t need it as target, and it is not an input feature. Must be dropped
- be careful of categorical values that appear only in the test data when you use the categorical encoder to transform them

Lab ToDo

1. process dataset into DataFrames for training and testing
   - split features/targets
   - encode categorical features and targets
   - normalize data based on train distribution
2. train traditional ML models
   - Decision Tree, RandomForest, others if you want (see scikit learn Estimators)
   - both binary and multiclass variants
3. train a DNN
   - create and train a dnn using Pytorch
   - both binary and multiclass variants
4. evaluate final performance
   - play around with different hyperparameters to see how things change
   - add dropout layers in the model to see if/what changes